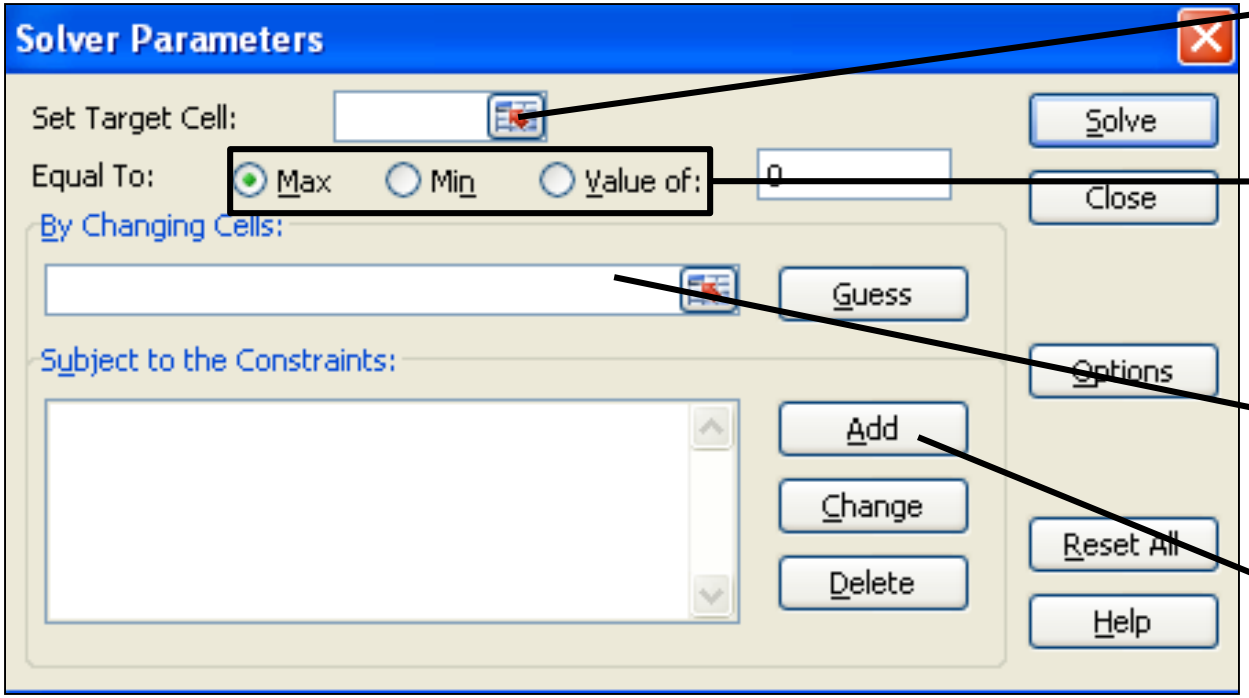
Lesson 8
Optimization

Excel in Business

Optimization In Excel

A business analyst often has to make recommendations that maximizes or minimizes a goal. In Excel this goal is achieved by using Solver tool in Excel.

SOLVER Tool In Excel



Cell corresponding to objective function

Specify the type of problem

Specify the variable cells

Add constraints

Example 1

Refer to worksheet ***LPP***

$$\text{max} \quad z = 2x + 3y$$

$$\text{s.t.} \quad 5x + 2y \le 19$$

$$13x - 6y \le 23$$

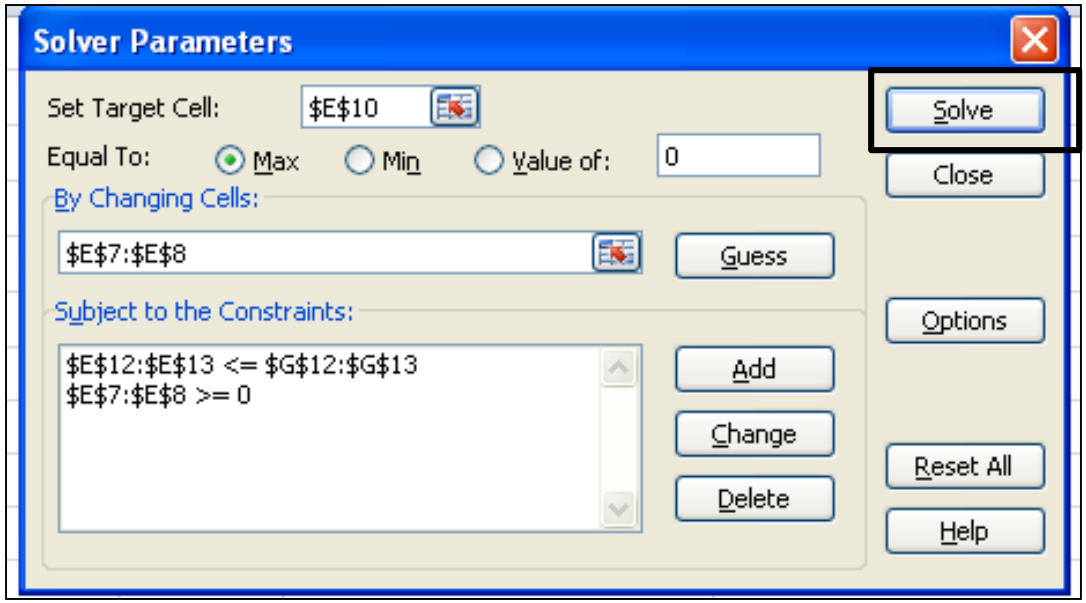
$$x, y \ge 0$$

Example 1 in Excel

	A	B	C	D	E	F	G
1							
2		max z = 2x+3y					
3							
4		5x+2y <= 19					
5		13x-6y <=23					
6		x,y>=0					
7				x	1		
8				y	1		
9							
10				z (2x+3y)	=2*E7+3*E8		
11							
12				5x+2y	=5*E7+2*E8	<=	19
13				13x-6y	=13*E7-6*E8	<=	23
14							

Initial guess

Example 1 in Excel (cont..)



Solution: x = 0, y = 9.5

Example 2

Example 2

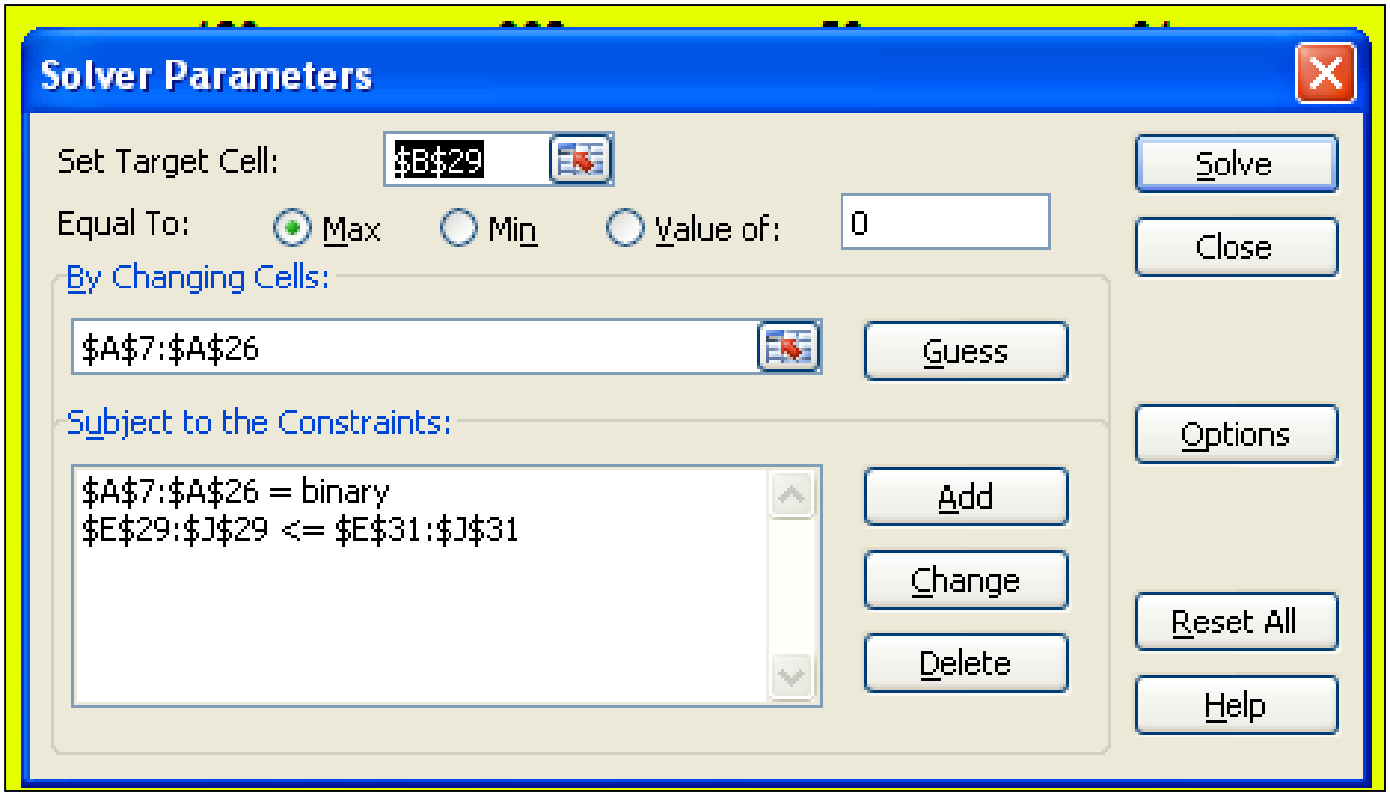
Refer to worksheet ***Project Choice***

TCS wants to start new projects to maximise their profit. They have 20 options in front of them. Due to financial and workforce constraints they cannot undertake all 20 projects. **What projects would you recommend to maximize their profit**?

Example 2 in Excel

	A	B	C	D	E	F	G	H	I	J
6	Do IT?		NPV		Cost Year 1	Cost Year 2	Cost Year	Labor Year	Labor Year	Labor Year
7	1	Project 1	928		398	180	368	111	108	123
8	1	Project 2	908		151	269	248	139	86	83
9	1	Project 3	801		129	189	308	56	61	23
10	1	Project 4	543		275	218	220	54	70	59
11	1	Project 5	944		291	252	228	123	141	70
12	1	Project 6	848		80	283	285	119	84	37
13	1	Project 7	545		203	220	77	54	44	42
14	1	Project 8	808		150	113	143	67	101	43
15	1	Project 9	638		282	141	160	37	55	64
16	1	Project 10	841		214	254	355	130	72	62
17	1	Project 11	664		224	271	130	51	79	58
18	1	Project 12	546		225	150	33	35	107	63
19	1	Project 13	699		101	218	272	43	90	71
20	1	Project 14	599		255	202	70	3	75	83
21	1	Project 15	903		228	351	240	60	93	80
22	1	Project 16	859		303	173	431	60	90	41
23	1	Project 17	748		133	427	220	59	40	39
24	1	Project 18	668		197	98	214	95	96	74
25	1	Project 19	888		313	278	291	66	75	74
26	1	Project 20	655		152	211	134	85	59	70
27										
28		Total NPV								
29				Used	=SUMPRODUCT($A$7:$A$26,E7:E26)	=SUMPRODUCT($A$7:$A$26,F7:F26)	=SUMPRODU	=SUMPRODU(	=SUMPROD	=SUMPRODU
30					<=	<=	<=	<=	<=	<=
31		=SUMPRODUCT(A7:A26,C7:C26)		Available	2500	2800	2900	900	900	900

Example 2 in Excel (cont..)



Solution: Total NPV = 9293

Example 3

Example 3

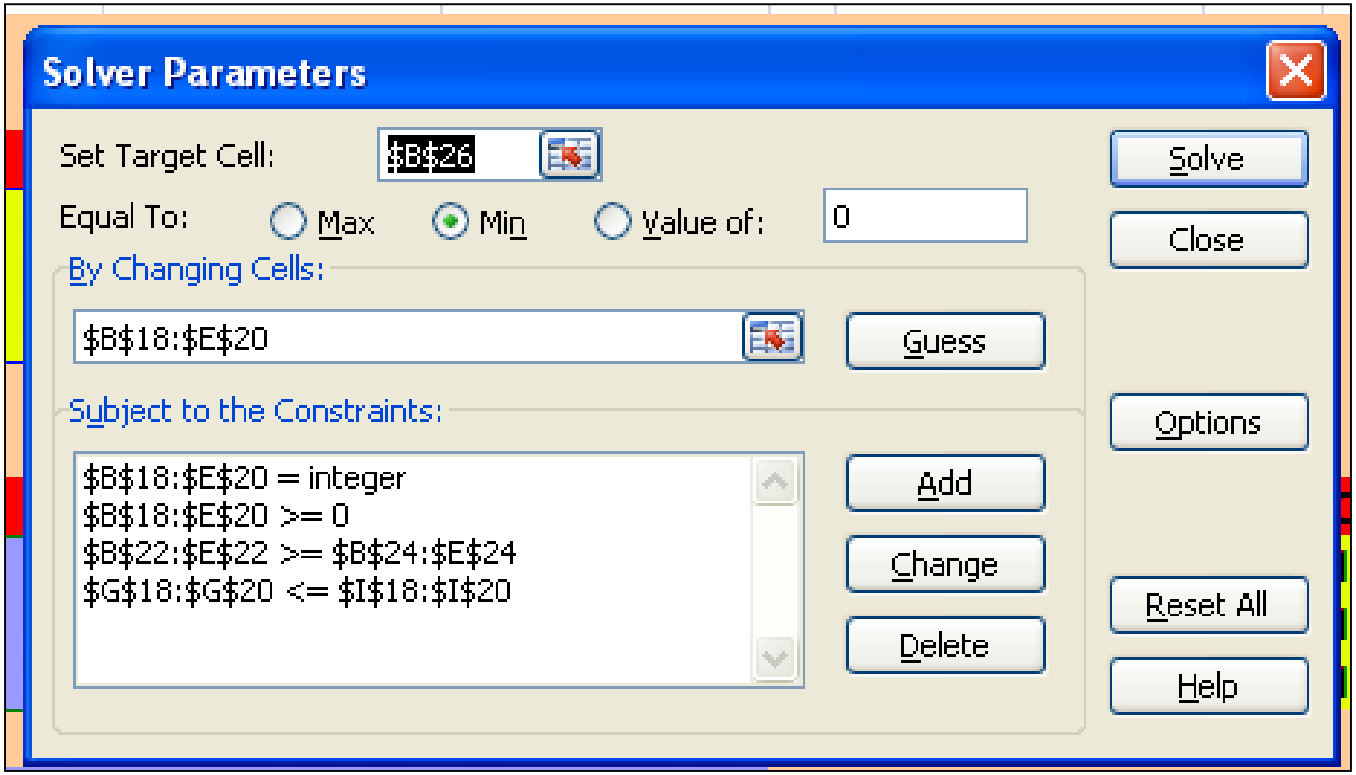
Refer to worksheet ***Transport***

A company produces a certain drug at its Los Angeles, Atlanta, and New York facilities. Each month, the Los Angeles plant can produce up to 10000 pounds of drug. Atlanta can produce up to 12000 pounds and New York can produce up to 14000 pounds. The company must make monthly shipments to the four regions of the United States -East, Midwest , South and West- the number of pounds are given below in Row8.The cost of shipment per pound, to each of these regions from the facilities is given in B12:E14. For example, the cost of shipping a pound of drug from LA to East is $5 and LA to West is $2.2. **Calculate the pounds to be shipped to different regions from the respective plants so that the transportation cost is minimum.**

Example 3 in Excel

	A	B	C	D	E	F	G	H	I
6	East is $5 anc								
7	Calculate the p								
8									
9	Cost of Shippir								
10									
11		EAST	MIDWEST	SOUTH	WEST				
12	LA	5	3.5	4.2	2.2				
13	ATLANTA	3.2	2.6	1.8	4.8				
14	NEW YORK	2.5	3.1	3.3	5.4				
15									
16	Pounds to be S								
17		EAST	MIDWEST	SOUTH	WEST		Sent		CAPACITY
18	LA	1	1	1	1		=SUM(B18:E18)	<=	10000
19	ATLANTA	1	1	1	1		=SUM(B19:E19)		12000
20	NEW YORK	1	1	1	1		=SUM(B20:E20)		14000
21									
22	Total	=SUM(B18:B20)	=SUM(C18:C20)	=SUM(D18:D20)	=SUM(E18:E20)				
23		>=	>=	>=	>=				
24	Demand	9000	6000	6000	13000				
25									
26	Total Cost	=SUMPRODUCT(B12:E14,B18:E20)							
27									

Example 3 in Excel (cont..)



Example 3 in Excel (solution)

F27

	A	B	C	D	E	F	G	H	I
6	East is $5 and LA to West is $2.2.								
7	Calculate the pounds to be shipped to different regions from the respective plants so that the transportation cost is minimum								
8									
9	Cost of Shipping (per pound)								
10									
11		EAST	MIDWEST	SOUTH	WEST				
12	LA	$5.00	$3.50	$4.20	$2.20				
13	ATLANTA	$3.20	$2.60	$1.80	$4.80				
14	NEW YORK	$2.50	$3.10	$3.30	$5.40				
15									
16	Pounds to be Shipped?								
17		EAST	MIDWEST	SOUTH	WEST		Sent		CAPACITY
18	LA	0	0	0	10000		10000	<=	10000
19	ATLANTA	0	3000	6000	3000		12000		12000
20	NEW YORK	9000	3000	0	0		12000		14000
21									
22	Total	9000	6000	6000	13000				
23		>=	>=	>=	>=				
24	Demand	9000	6000	6000	13000				
25									
26	Total Cost	86800							
27									